

SAI DHRUV YELLANKI

College Park, Maryland (Ready to Relocate)

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📁 [Portfolio](#)

Professional Summary

AI & Machine Learning Engineer with 3+ years of experience architecting production-grade Generative AI systems and predictive pipelines. Currently specialized in multi-agent workflows and RAG orchestration using LangGraph, LangChain, and vLLM. Expert in LLM fine-tuning (LoRA, RLHF) and scaling LLMOps via Kubernetes and Azure Databricks to deliver resilient, enterprise-scale deployments across healthcare, finance, and supply chain domains.

Education

University of Maryland

Master of Science in Data Science; GPA: 3.81/4.0

August 2024 – May 2026

College Park, MD, USA

Experience

Aurelius Tech & Talent Solutions

AI Engineer Intern

June 2025 – Present

Ellicott City, MD, USA

- Architected an automated proposal generation system using an LLM-based RAG pipeline by integrating Azure OpenAI text-embedding-3 and pgvector, reducing document retrieval time by 65% while maintaining a 500ms latency.
- Orchestrated a multi-agent Chain-of-Thought framework to drive the proposal system by fine-tuning a GPT-OSS-20B orchestrator to coordinate GPT-4 agents, achieving 79% precision and 74% recall in compliance detection.
- Streamlined production LLMOps workflows for the proposal platform using Docker and Kubernetes to develop containerized CI/CD pipelines, shortening release cycles by 40% and ensuring 95% system uptime.
- Optimized document processing throughput by 35% for the proposal system by implementing Redis semantic caching and distributed processing to integrate unstructured data in real-time.

Watzmann Consulting

Data Analyst – Predictive Analytics & Machine Learning

June 2023 – June 2024

Hyderabad, TG, India

- Developed a high-performance demand forecasting ensemble using XGBoost and Bi-LSTM models for 200K+ records, improving sales prediction accuracy by 12% (MAE less than 0.6) to optimize inventory levels and reduce staffing overhead.
- Architected a workload optimization system that paired Bi-LSTM runtime forecasts with Mixed-Integer Linear Programming (MILP), reducing idle compute time by 18% and significantly cutting infrastructure costs.
- Operationalized end-to-end model retraining and deployment by implementing MLOps pipelines with MLflow and Azure Machine Learning, boosting model reliability and operational efficiency by 15% using real-time Power BI monitoring.

Ramphal Technologies

Data Science Intern

December 2022 – May 2023

Hyderabad, TG, India

- Trained system-performance prediction models using PyTorch and XGBoost to forecast application latency and server load, improving accuracy by 17% and reducing average request delay by 7.5ms.
- Constructed a real-time event pipeline with RabbitMQ and PostgreSQL to process 10,000+ telemetry events per second, cutting reporting latency by 30% via real-time Tableau dashboards.

Projects

LoRA-Tuned RLHF News Summarization System | RLHF, RAG, LORA, Langchain, RAG, Docker, vLLM | [GitHub Link](#)

- Deployed a news-summarization platform using RLHF (PPO/trlx) to fine-tune a T5 model integrated with a pgvector RAG pipeline, improving ROUGE-L by 11% and achieving 240ms p95 latency using vLLM.

BioGraphX: Multi-Agent Biomedical RAG System | Python, LangGraph, RAG, Qwen2.5, ChromaDB, Docker | [GitHub Link](#)

- Architected a production-ready biomedical QA system by orchestrating 7 specialized LangGraph agents and a LoRA fine-tuned Qwen2.5 model, achieving a 96% ROUGE-L improvement and enabling scalable Docker/FastAPI deployment with sub-8s inference over 298K PubMed records.

Technical Skills

Programming Languages & Databases: Python (Pandas, NumPy, Scikit-learn, PySpark, TensorFlow, PyTorch, SpaCy, NLTK, Matplotlib, Seaborn, Plotly, Dash), SQL, R, PostgreSQL, Redis, MySQL, MongoDB.

Tools & Technologies: Docker, Kubernetes, Azure (Machine Learning, DevOps, Databricks), AWS (Bedrock, Sagemaker), MLflow, FastAPI, vLLM, Git, Power BI, Tableau, Grafana, CI/CD Pipelines, Microsoft Copilot, Microsoft Power Platform.

Machine Learning & Analytics: Predictive Modeling, Deep Learning, Time Series Forecasting (Prophet, ARIMA), MLOps, model stacking, Model Deployment & Monitoring, Anomaly Detection, hyperparameter tuning, cross-validation.

AI & LLM Engineering: NLP (NER, summarization, sentiment analysis, embeddings, transformers), fine-tuning (LoRA, QLoRA, RLHF), RAG, multi-agent orchestration (LangChain, LlamaIndex, LangGraph, AutoGen), Prompt Engineering, inference (vLLM).